



# Air Quality Situation Report

Edmonton Symphony Orchestra Outdoor Concert · Hawrelak Park (Heritage Amphitheatre)

Sat, Sep 5, 2026 · 12:37 AM MDT

## Air Quality Health Index (AQHI) — Edmonton

# 1.0

Current AQHI at Edmonton McCauley (4.75 km from venue)  
Forecast Sat 3:00 AM at Edmonton McCauley (4.75 km from venue): **1.0**

| Health Risk | AQHI | At-Risk Population                   | General Population                        |
|-------------|------|--------------------------------------|-------------------------------------------|
| Low Risk    | 1.0  | Enjoy your usual outdoor activities. | Ideal air quality for outdoor activities. |

No active Environment Canada weather alerts for the venue.

## Environmental Intelligence Summary

- At Edmonton Symphony Orchestra Outdoor Concert, temperature is 12.6°C and feels near 11.3°C. Winds are 12 km/h from NW, gusting near 26 km/h.
- The nearest current AQHI is 1 at Edmonton McCauley, 4.8 km from the venue.
- The official Government of Alberta AQHI for Edmonton is 1 (forecast tonight: 2, tomorrow: 2).
- A gridded AQHI estimate blending official and community sensors (confidence: High, nearest contributing point 4.0 km away) puts the area near 1.
- The nearest air monitoring station (Edmonton Lendrum, 3.5 km) reports fine particulate matter (PM2.5) at 2.7 µg/m³.
- A nearby community sensor ('ACA\_LaurierHeightsCL', 1.8 km) reads 1.0 µg/m³ PM2.5.
- Back-trajectory modeling (HRDPS wind fields, 6 h lookback) indicates the air currently at the venue traveled from the vicinity of a point 149.3 km to the N, i.e. arriving from the N.
- The nearest active fire detection (NASA FIRMS – VIIRS, last 2026-09-04) is 123.8 km SW of the venue. The modeled back-trajectory does not pass near this fire (nearest approach 244.2 km), so this specific fire is unlikely to be the source of any smoke reaching the venue right now.
- The AQHI forecast for Sat 3:00 AM is 1, based on Edmonton McCauley, 4.8 km from the venue.
- Precipitation probability reaches approximately 53%.
- Overall operational environmental risk is MODERATE.

## Recommendations

- Environment Canada AQHI guidance — general population: Ideal air quality for outdoor activities.
- Environment Canada AQHI guidance — at-risk populations: Enjoy your usual outdoor activities.

## Hazard Snapshot — Overall Risk: Moderate Risk

|                                                                |                                                     |                                                |                                                                     |                                                            |
|----------------------------------------------------------------|-----------------------------------------------------|------------------------------------------------|---------------------------------------------------------------------|------------------------------------------------------------|
| AIR QUALITY<br><b>Low Risk</b><br>1.0 AQHI                     | HEAT<br><b>Low Risk</b><br>14.6 °C apparent/humidex | WIND<br><b>Low Risk</b><br>27.0 km/h peak gust | PRECIPITATION<br><b>Moderate Risk</b><br>0.4 mm/h maximum           | THUNDERSTORM<br><b>Low Risk</b><br>— first forecast signal |
| AQHI RATE OF CHANGE<br><b>Low Risk</b><br>0.0 AQHI change/hour |                                                     |                                                | WIND SHEAR<br><b>Low Risk</b><br>8.2 ° direction diff (10m vs 120m) |                                                            |

## Local Monitoring

|                                                              |                                                   |                                                                   |                                                                        |
|--------------------------------------------------------------|---------------------------------------------------|-------------------------------------------------------------------|------------------------------------------------------------------------|
| Nearest station<br><b>1.0</b><br>Edmonton McCauley · 4.75 km | Blended estimate<br><b>1.0</b><br>confidence High | PM2.5 (station)<br><b>2.7 µg/m³</b><br>Edmonton Lendrum · 3.51 km | PM2.5 (community)<br><b>1.0 µg/m³</b><br>ACA_LaurierHeightsCL · 1.8 km |
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Prepared by Alberta Capital Airshed for Edmonton Symphony Orchestra outdoor concerts. Beta decision-support product  
— for the interactive map, live cameras and hourly outlook, scan the QR code or visit [https://dkevinm.github.io/](https://dkevinm.github.io/eso_sitre/)  
eso\_sitre/

